Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications

Diversity Combination Techniques

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Diversity combining

- All diversity techniques rely on the provision of two or more channels between which the fading correlation is low (ideally zero)
- Diversity channel outputs are combined by
  - Switched. With 4 inputs ??
  - Select
  - Sum them together EGC Equal Gain Combination
  - Sum the best ones. Maximum Ratio Combination

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Diversity combining performance

- Assume the following (for all diversity combining techniques):
  - ≥ 2 – N branches (diversity order N)
  - Uncorrelated fading in each branch Actually $\simeq$ 70%
  - SNR (as a ratio – not in dB) of i^{th} channel is γ_i
  - SNR (as a ratio – not in dB) after applying N-channel diversity is $\gamma_{div,N}$
  - All channels subject to equal noise power

improves due to averaging & noise suppression

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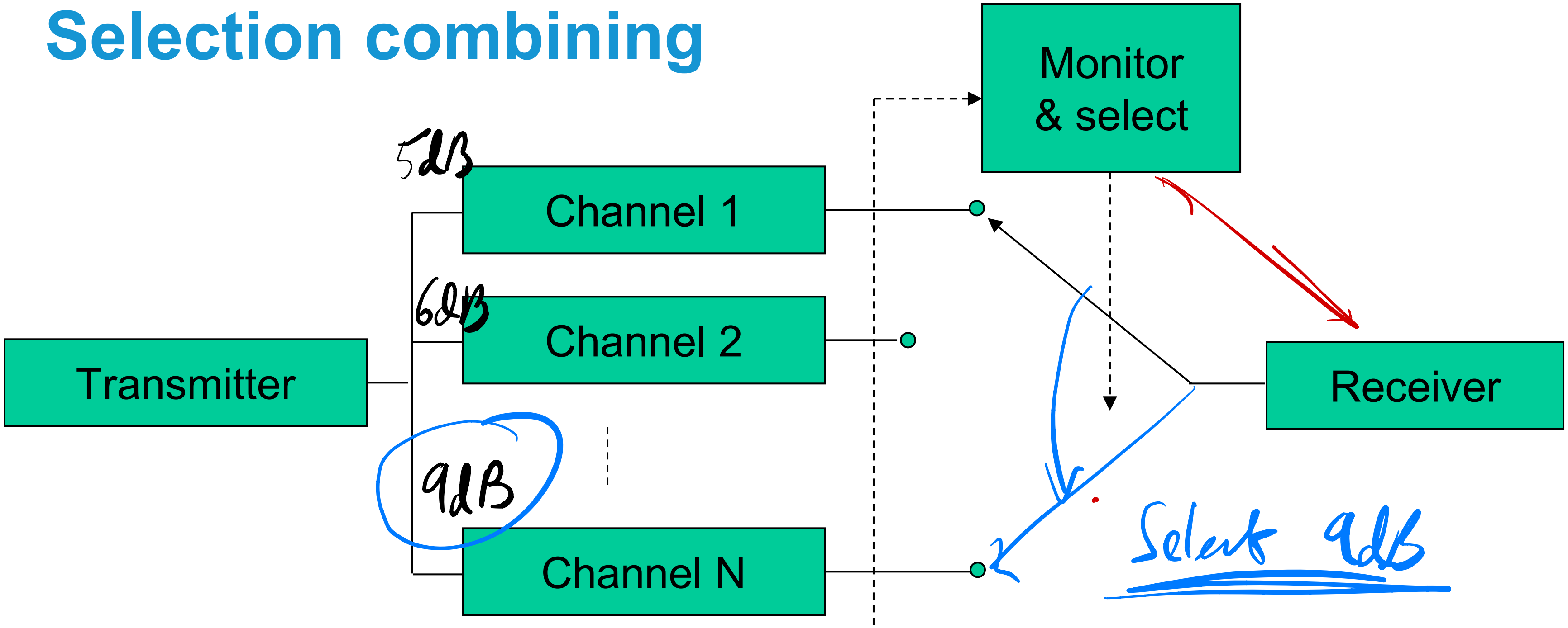
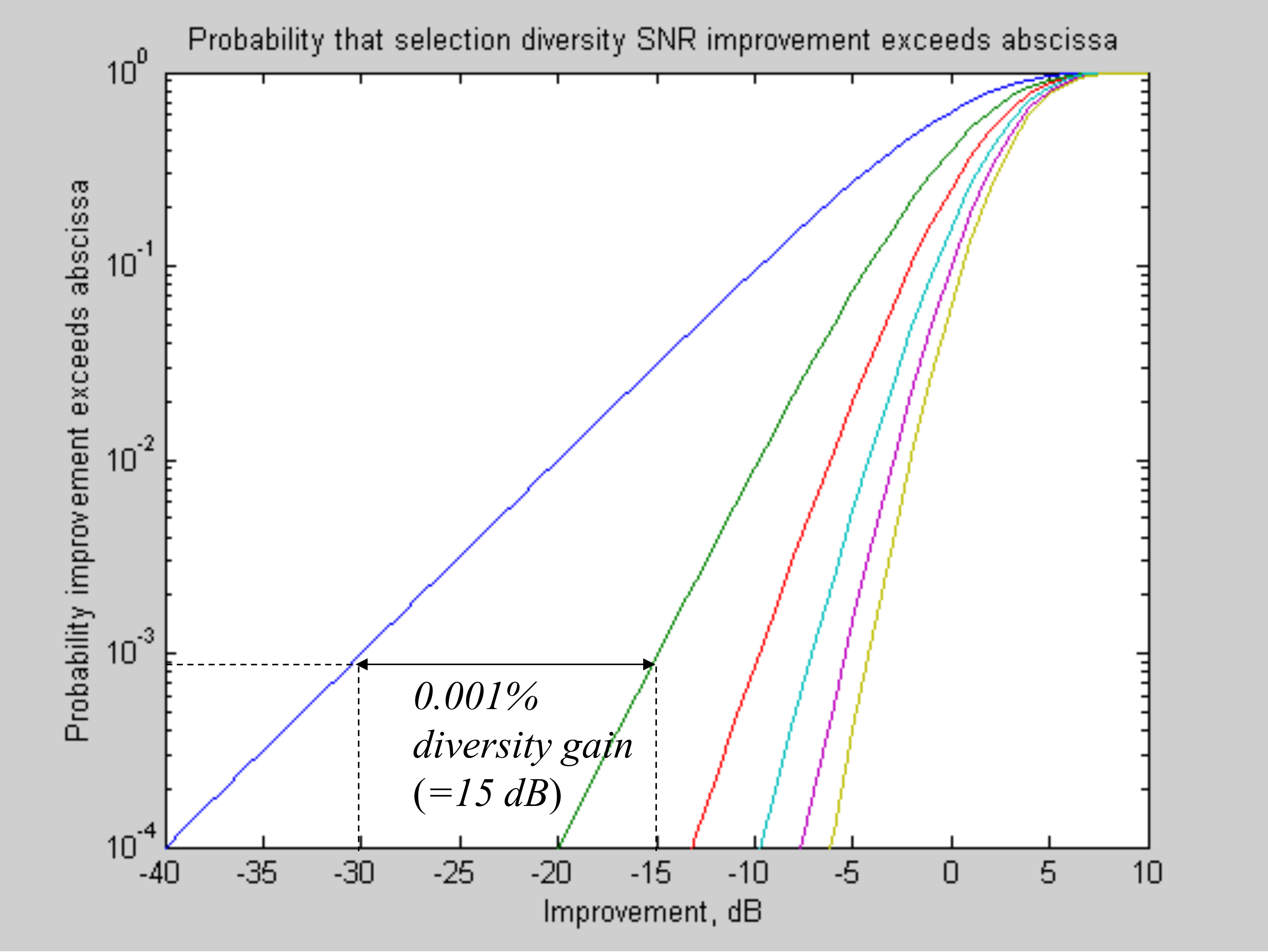
Selection combining



- Channels are continuously monitored and the one with the best SNR is selected
- Ensures channel with best SNR always used
- Needs a (simplified) receiver for each channel to allow continuous monitoring of all SNRs



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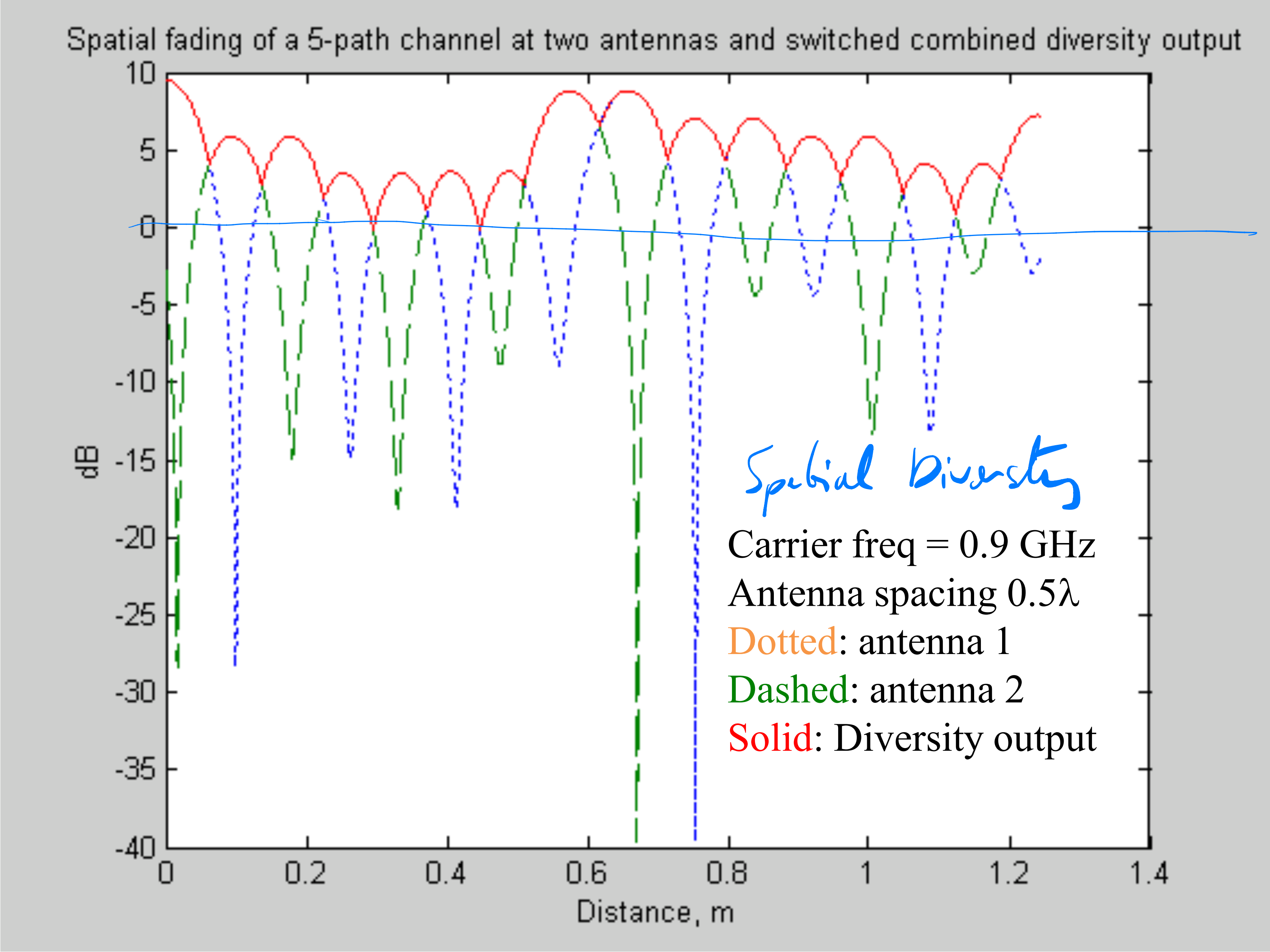

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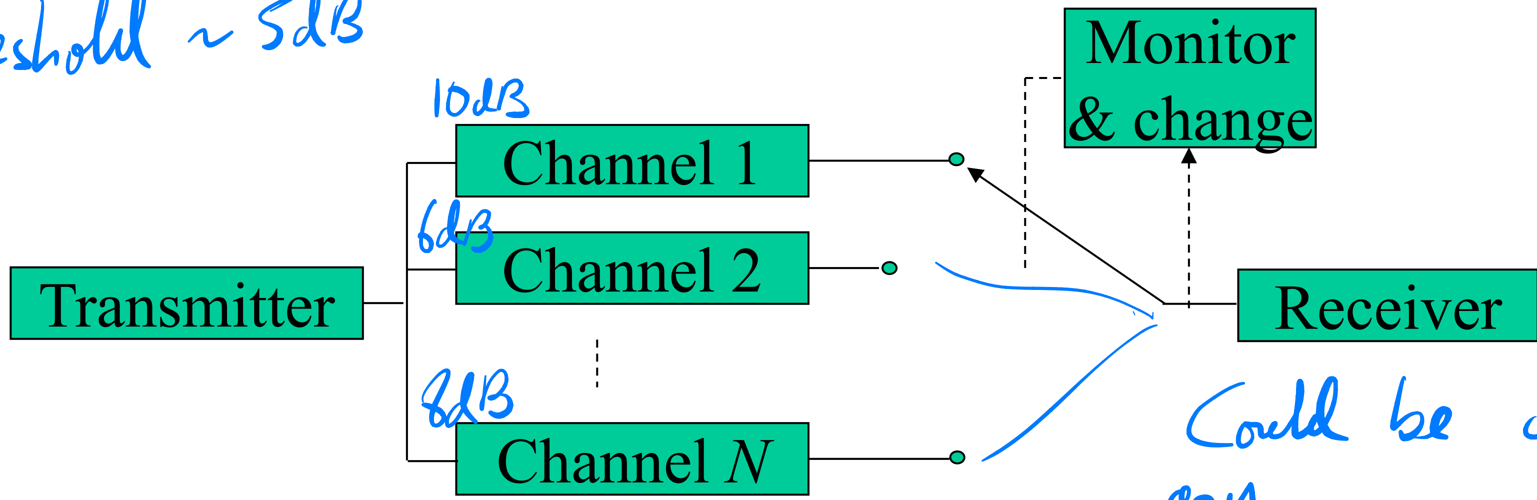
Switched Combining

Threshold ~ 5dB

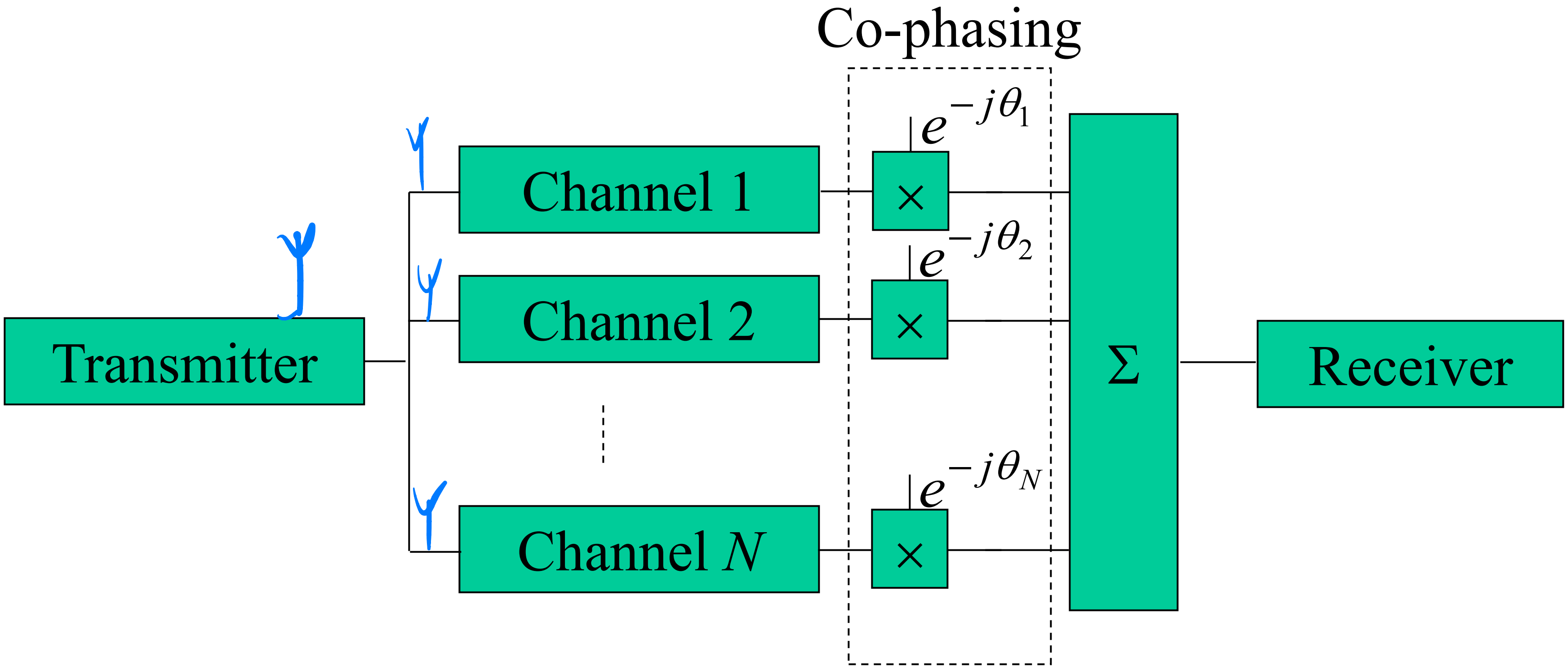


Could be on any.

- A channel remains selected until its SNR drops below a specified threshold
- Each channel is then tried in turn until one is found with SNR above specified threshold
- Switched diversity therefore guarantees only that an acceptable (rather than the best) channel is used - assuming an acceptable channel exists!
- Only requires a single receiver - simpler than selection combining

While simple work very well. see graph

Equal Gain Combining (EGC)



- Outputs of all channels are co-phased and summed

Shifts phase of diverse signal to sync & sums them for a strong signal.

Equal Gain Combining performance

- The signals in all branches are co-phased and then summed
- All power received in all branches is therefore utilised
  - In contrast to selection and switched combing where at any instant of time signal power in all channels but one is lost
- Clearly an improved SNR results since
  - Co-phased signals add voltage-wise
  - Incoherent noise adds power-wise
- Performance is thus better than selection diversity

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Equal Gain Combining performance

- For 2-channel equal gain combining the post diversity SNR is given by [Saunders S, 1999]:

$$\gamma_{div,2} = \frac{\gamma_1 + \gamma_2 + 2\sqrt{\gamma_1 \gamma_2}}{2}$$

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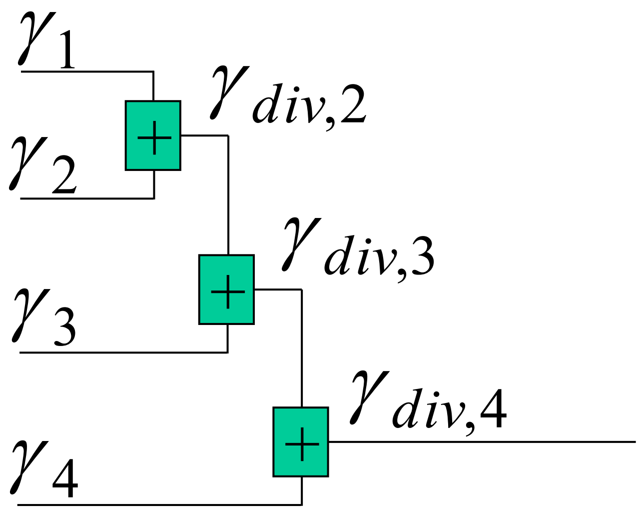
Equal Gain Combining performance

- Summing all co-phased channels is equivalent to adding first two, then adding third to the sum then adding forth to the sum etc. as illustrate below



- The 2-channel SNR formula can therefore be extended to *N* channels using:

$$\gamma_{div,N} = \frac{\gamma_{div,N-1} + \gamma_N + 2\sqrt{\gamma_{div,N-1}\gamma_N}}{2}$$

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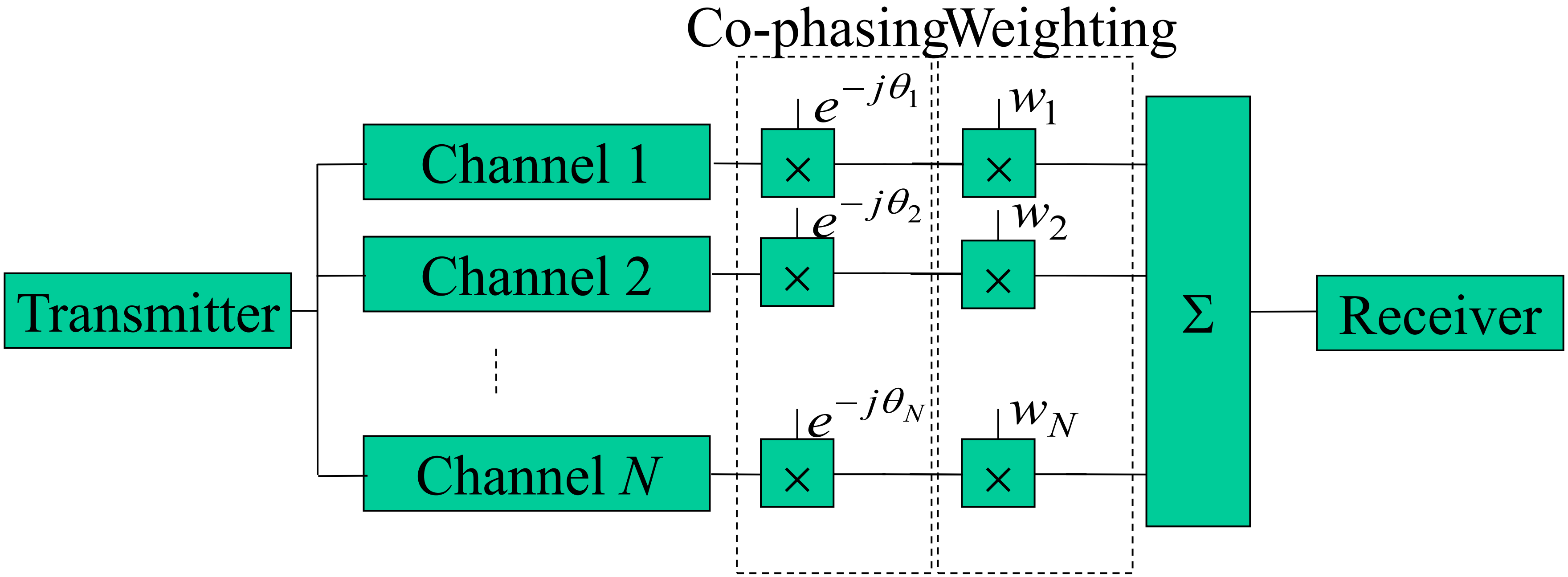
Maximum Ratio Combining (MRC)



- Outputs of channels are co-phased and then weighted in proportion to their SNRs before being summed
- The co-phasing and weighting can be combined in one *complex* weighting process

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Maximum ratio combining performance

- Channels are co-phased and weighted in proportion to their SNRs before summing
- This gives best possible diversity combining performance in which SNR after diversity combining is the sum of the SNRs of each channel, i.e.:

$$\gamma_{div,N} = \sum_{i=1}^{N} \gamma_i$$

Example:

Output of switched = Any - whichever it lands on 1st.
Selection = 10dB. Picks the best.

- The SNRs at the output of three spatially separated antennas are 6 dB, 8 dB and 10 dB. What is the SNR at the output of an ideal maximal ratio combiner? What is the SNR at the output of an ideal equal gain diversity combiner? Assuming all are above threshold.
- - First convert the given SNRs from dBs to ratios:

Divide by 10
inv log.

$$SNR\_1 = 10^{\frac{6}{10}} = 3.98, \quad SNR\_2 = 10^{\frac{8}{10}} = 6.3, \quad SNR\_3 = 10$$

For maximum ratio combining:

$$\left.\frac{S}{N}\right|_{MRC} = \sum_{i=1}^{N}\left(\frac{S}{N}\right)_i = 3.98 + 6.3 + 10 = 20.28 = 10\log_{10} 20.28 \ \text{dB} = 13.1 \ \text{dB}$$

- - For equal gain combining:

$$\gamma_{div,N} = \frac{\gamma_{div,N-1} + \gamma_N + 2\sqrt{\gamma_{div,N-1}\gamma_N}}{2}$$

1 & 2

$$\gamma_{div,2} = \frac{\gamma_{div,1} + \gamma_2 + 2\sqrt{\gamma_{div,1}\gamma_2}}{2} = \frac{3.981 + 6.310 + 2\sqrt{3.981 \times 6.310}}{2} = 10.16 = 10.1 \ \text{dB}$$

3

$$\gamma_{div,3} = \frac{\gamma_{div,2} + \gamma_3 + 2\sqrt{\gamma_{div,2}\gamma_3}}{2} = \frac{10.16 + 10 + 2\sqrt{10.16 \times 10}}{2} = 20.16 = 13.0 \ \text{dB}$$

" $10\log 20.16$

Comparison of combining techniques in practice

30%

- Although uncorrelated diversity channels are often assumed for analysis purposes, good diversity advantage is realised even if fading in channels is partially correlated
- Although the rank order of combining techniques is MRC, EGC, Selection, Switched, the performance differences between them are small compared with the performance improvement
- The simpler techniques (e.g. switched and selection combining) are therefore most often used in practice

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Summary 1

- Various forms of diversity can be applied to improve fading performance including:
  - Space diversity
  - Polarisation diversity
  - Time diversity
  - Frequency diversity
  - Angle diversity
- Space diversity is the most frequently applied diversity technique in practise
- The diversity spacing required at a mobile terminal in a richly scattering environment is typically half a wavelength

$\lambda/2$

Remember our linear arrays as well!

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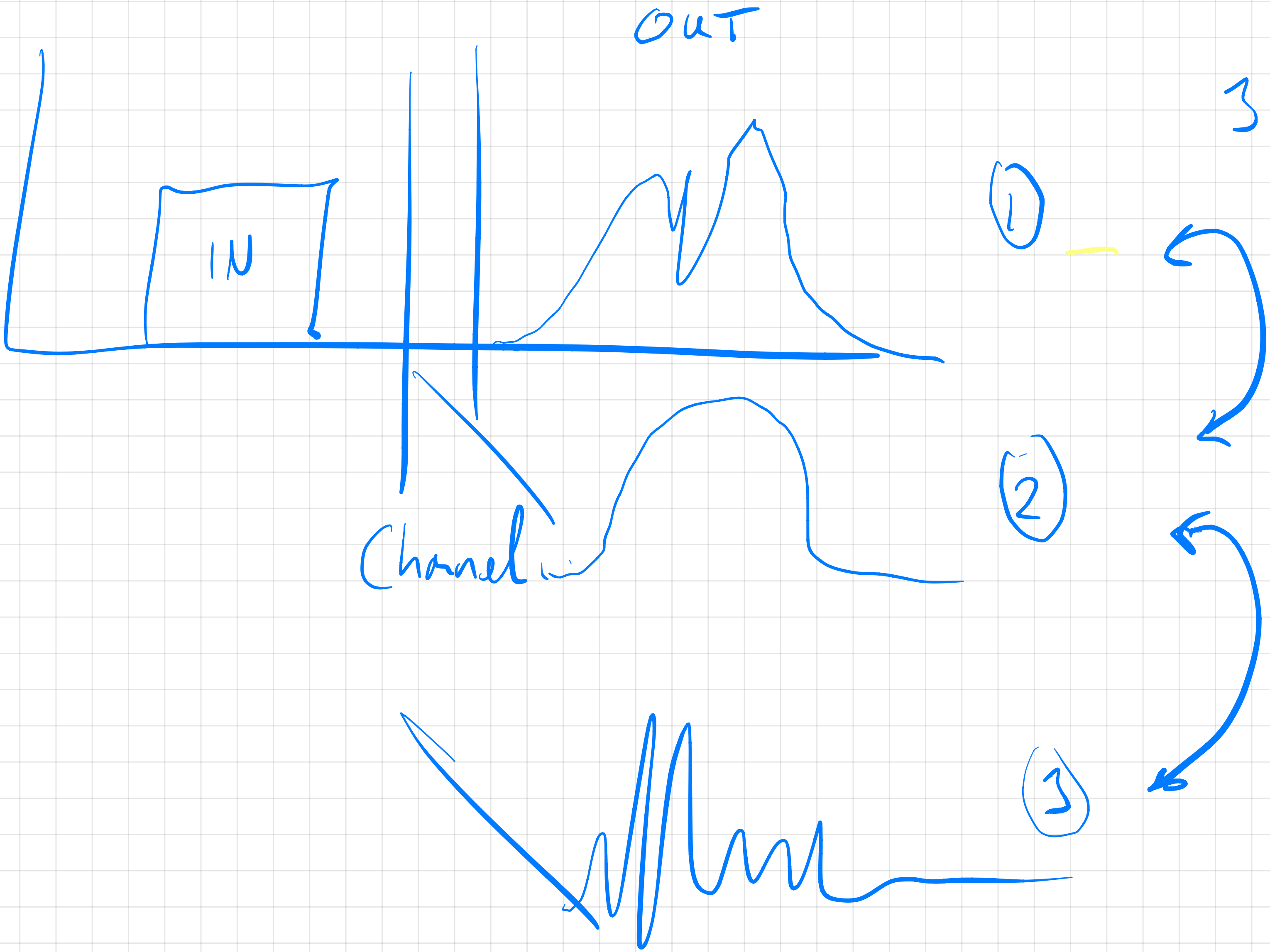
Summary 1

- The diversity spacing required at a basestation is typically many wavelengths
- Various forms of diversity combining technique can be applied including (in increasing order of performance):
  - Switched combining
  - Selection combining
  - Equal gain combining
  - Maximum ratio combining

Increasing performance

- The difference in performance between the best and worst diversity techniques is small ~ 3dB ~ double !!
- Switched or selection combining is usually used in practice
- Almost full diversity advantage is obtained with channel correlation of around 70%

3 Diverse channels



Part correlated

Decorrelated

Correlated - matching or similar

De - - not matching